

Pranay Meshram

Buffalo, NY

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🐙 GitHub

Education

University at Buffalo, The State University of New York

PhD in Computer Science and Engineering

Present

University at Buffalo, The State University of New York

Master of Science: Engineering Science Robotics

Dec 2022

Technical Skills

Programming Languages: C, C++, Python, Rust

Frameworks & Tools: PyTorch, NumPy, SciPy, Pandas, OpenCV, ROS 2, FastAPI, Ollama, Gazebo, NVIDIA Isaac Sim

Robotics & AI: Path Planning, Robot Learning, Hierarchical Abstraction, Hypergraphs, Language-Conditioned & Multimodal Learning, RL

Perception: Computer Vision, Stereo Vision, Depth Estimation, 3D Reconstruction, Point-Cloud Processing

Research Experience

University at Buffalo, The State University of New York

Research Assistant

Present

Language-Conditioned Robot-Adaptive Off-Road Path Planning

- Introduced language-conditioned planning that adapts off-road routes to mission preferences under robot-feasible constraints.
- Built Torch/CUDA cross-attention grounding LLM intent in semantic terrain and robot-specific traversal consequences, enabling adaptation across wheeled and legged robots.
- Developed a two-stage region-guided planner that reduced search states by 75% and candidate latency by 20–22% versus dense search while preserving feasibility across tested maps and robot models.

HOPHY: Hierarchical Off-Road Planning using Hypergraphs

- Developed HOPHY, a hierarchical abstract representation of unstructured off-road terrain, enabling $68\text{--}1,280\times$ faster path queries than pixel-level A* with 0.00% median cost difference.
- Represented localized environmental changes as hyperedge cost updates, enabling fast replanning without rebuilding the spatial graph (67.5 ms update; 137.1 ms replacement query).
- Validated HOPHY through 1.5 km of autonomous Clearpath Jackal navigation in unstructured terrain.

CLEAR: Semantic-Geometric Terrain Abstraction

- Developed CLEAR (Connected Landcover Elevation Abstract Representation), fusing dense elevation and semantic land-cover data to accurately estimate physics-based traversability, enabling large-scale autonomous navigation.
- Demonstrated $10\times$ faster planning with only 6.7% cost overhead, yielding 6–9% shorter and significantly more reliable paths compared to existing grid and mesh-based baseline abstractions.

Quality-Aware 3D Reconstruction

- Designed Quality-Aware Loss (QAL) decoupling precision and recall, improving coverage by +4.3 points and recovering fine structures.
- Created a 4D quality metric for point-clouds comprehensively measuring 3D reconstruction coverage, precision, and consistency.

Meta, Reality Labs

Redmond, WA

Research Scientist Intern

May 2024 - Sept 2024

Self-supervised Stereo Depth Estimation from Polarization

- Built a self-supervised stereo depth pipeline leveraging polarization physics and feature representations to enhance geometric consistency in texture-less and reflective regions.
- Designed a multi-objective learning framework jointly predicting depth and surface normals, achieving a 20% accuracy improvement over state-of-the-art stereo models.

Publications

- HOPHY: Hierarchical Off-Road Planning using Hypergraphs. Under submission.
- CLEAR: A Semantic-Geometric Terrain Abstraction for Large-Scale Unstructured Environments. **Accepted**, IEEE RA-L, Aug. 2026.
- QAL: A Loss for Recall-Precision Balance in 3D Reconstruction. **Published**, WACV 2026.
- Empir3D: Point-Cloud Quality Assessment Framework. **Published**, CRV 2026.

Professional Experience

NVIDIA Graphics Pvt Ltd

Pune, India

System Software Engineer

Dec 2018 – May 2021

- Led development of CUDA language program generating Fuzzer using Deep Learning and implemented solutions reducing engineering effort by 30%.
- Managed the design and verification of the CUDA Compiler for Automotive Safety Project to satisfy ISO 26262.

Renishaw Metrology Systems Ltd

Pune, India

Software Engineer

Jul 2017 – Dec 2018

- Implemented voxel-based motion path planning and probe compensation algorithms for precision metrology.
- Applied Particle Swarm Optimization to calibration routines → 25% reduction in recurring bug count.

Accomplishments and Achievements

2022: 1st, ACM/IEEE TinyML Design Contest at ICCAD; 4th, DAC System Design Contest.